AGREEMENT SUBTREES OF BALANCED TREES: THE EXPONENT EXISTS AND LIES BETWEEN 0.243 AND 5/11

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Abstract. Let $M(n)$ be the smallest possible size of a maximum agreement subtree of two balanced rooted binary phylogenetic trees on the same $n = 2^m$ leaves. Martin and Thatte conjectured $M(n) \geq n^{1/2}$. Bordewich, Linz, Owen, St. John, Semple and Wicke disproved this, showing $M(n) < c\, n^{1/2}$ for every $c > 0$ along a subsequence, and proved $M(n) \geq n^{0.17}$. A substitution lemma, a form of the tree-product argument of Guillemot, Nicolas, Berry and Paul, shows that M is submultiplicative, so the exponent $\beta = \lim \log M(n)/\log n$ exists and equals $\inf_m \log_2 M(2^m)/m$. The example of Bordewich et al. on 2048 leaves then gives $\beta \leq 5/11$, that is, $M(n) = O(n^{5/11})$. Optimising the parameters in their inductive lower bound gives $M(n) \geq n^{0.235}$, and an induction that looks two levels down one tree and one level down the other gives $M(n) \geq n^{0.243}$. Both steps rest on inequalities verified by computer in exact and interval arithmetic. Hence $0.243 \leq \beta \leq 5/11 \approx 0.4545$. All of this except the last inequality is also checked in Lean.

1. Introduction

Maximum agreement subtrees measure how much two phylogenetic trees on the same species have in common. The extremal question asks how small the largest common restriction can be. For unrooted binary trees the answer is $\Theta(\log n)$ [8], after earlier bounds by Steel and Székely [12] and by Martin and Thatte [10]. For rooted trees without a shape constraint the question is trivial, since a caterpillar and its reverse agree on only two leaves. Constraining the shape changes the picture. When both trees are *balanced*, Martin and Thatte proved a polynomial lower bound n^{β_0} with $\beta_0 \approx 0.149$ and conjectured that $n^{1/2}$ is the truth [10, Conjecture 20].

Bordewich, Linz, Owen, St. John, Semple and Wicke [2] gave a family of counterexamples. For every $k \geq 1$ they construct balanced trees on $2^{2^{k+1}-k-2}$ leaves whose maximum agreement subtrees have 2^{2^k-k} leaves, which is below $c\, n^{1/2}$ once k is large. They also raised the lower bound to $n^{0.17}$. In their family the ratio of the agreement size to $n^{1/2}$ tends to zero only like $(\log n)^{-1/2}$, so it leaves open whether the true growth rate is $n^{1/2-o(1)}$.

For a small case, the balanced trees $((1,2),(3,4))$ and $((1,3),(2,4))$ agree on no three leaves (Figure 1(a)), so $M(4) = 2 = 4^{1/2}$; for $k = 3$ the construction of [2] gives two balanced trees on 2048 leaves that agree on at most 32, against $2048^{1/2} \approx 45$. This note settles that point and narrows the range for the exponent from both sides.

Theorem 1.1. *Let $M(n)$ denote the minimum of* $\operatorname{mast}(S,T)$ *over all pairs of balanced rooted binary phylogenetic trees S, T on a common set of $n = 2^m$ labels.*

1. *The limit $\beta = \lim_{m\to\infty} \log_2 M(2^m)/m$ exists and equals $\inf_{m\geq 1} \log_2 M(2^m)/m$.*
2. *$M(n) \leq 2^{10}\, n^{5/11}$ for every $n = 2^m$. In particular $\beta \leq 5/11$.*
3. *$M(n) \geq n^{0.243}$ for every $n = 2^m$. In particular $\beta \geq 0.243$.*

Part (1) rests on a substitution lemma (Lemma 3.1). Its argument is the tree-product argument of Guillemot, Nicolas, Berry and Paul [5, Section 5.3], used there to show that the size of a maximum compatible tree, which equals mast for binary trees, is multiplicative under powers of a tree (a construction they cite to earlier work, including Hein, Jiang, Wang and Zhang [6]); we allow the substituted pair to vary from leaf to leaf, and apply it to balanced trees. Its useful consequence is that any single pair of balanced trees with a small agreement subtree yields a bound for all n. Part (2) applies it to the $k = 3$ example of [2]. Part (3) keeps the induction of [2, Theorem 1.3] but replaces its case analysis by an exact inequality in four variables, checked in interval arithmetic (Lemma 5.1), which gives $n^{0.235}$. Looking two levels down one tree and

one level down the other turns this into an inequality in eight variables between finitely many concave functions, which a certified simplicial subdivision verifies (Lemma 5.3); this gives $n^{0.243}$.

For comparison, two independent uniformly random leaf-labelled binary trees have a maximum agreement subtree of order at least $n^{0.366}$ [1], improved to $n^{0.4464}$ [7], and typically at most $n^{1/2-\varepsilon}$ [3]. For two balanced trees with independent uniform labellings the expected size is $\Theta(n^{1/2})$ [11]. Part (2) therefore shows that suitably chosen balanced trees agree polynomially less than random labellings of the same shapes.

2. Preliminaries

We follow the terminology of [2]. A rooted binary phylogenetic X-tree is a rooted binary tree whose leaves are bijectively labelled by X. For $Y \subseteq X$ the restriction $T|Y$ is obtained from the minimal subtree connecting Y by suppressing non-root vertices of degree two; for example, restricting $((1,2),(3,4))$ to $\{1,2,3\}$ gives $((1,2),3)$. For trees S and T with label sets X and X', a set $Y \subseteq X \cap X'$ is an *agreement set* if $S|Y \cong T|Y$ as leaf-labelled trees, and $\mathrm{mast}(S,T)$ is the largest size of an agreement set. A rooted binary tree is *balanced* if it has 2^m leaves, all at depth m. If S_L, S_R and T_L, T_R are the maximal pendant subtrees of S and T, then

$$\begin{aligned}\mathrm{mast}(S,T) = \max\{\, &\mathrm{mast}(S_L,T_L) + \mathrm{mast}(S_R,T_R),\ \mathrm{mast}(S_L,T_R) + \mathrm{mast}(S_R,T_L),\\ &\mathrm{mast}(S,T_L),\ \mathrm{mast}(S,T_R),\ \mathrm{mast}(S_L,T),\ \mathrm{mast}(S_R,T)\}.\end{aligned} \tag{1}$$

In words, a largest agreement set either meets both halves of both trees, matching the halves in one of the two possible ways, or lies inside one half of one of the trees.

For example, the balanced trees $S = ((1,2),(3,4))$ and $T = ((1,3),(2,4))$ of Figure 1(a) agree on no three leaves: any three labels contain exactly one cherry of S and exactly one cherry of T, and the cherries $\{1,2\},\{3,4\}$ of S never coincide with the cherries $\{1,3\},\{2,4\}$ of T. So $M(4) = 2 = 4^{1/2}$.

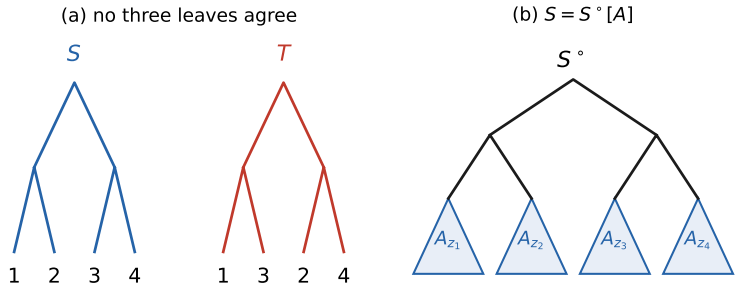


Figure 1. (a) Two balanced trees on four leaves whose largest agreement subtree has two leaves. (b) The substitution $S^\circ[A]$ of Section 3: each leaf z of S° becomes the root of a tree A_z. An agreement set of $S^\circ[A]$ and $T^\circ[B]$ can use at most $\mathrm{mast}(S^\circ,T^\circ)$ of the blocks, and at most $\mathrm{mast}(A_z,B_z)$ leaves inside each.

A *cluster* of a rooted tree is the set of leaves below some vertex; $\{1,2\}$ is a cluster of $((1,2),(3,4))$ and $\{2,3\}$ is not. If C is a cluster of S and Y meets C in exactly one leaf y, then for any $c \in C$ the tree $S|Y$ is obtained from $S|((Y \setminus C) \cup \{c\})$ by renaming c as y.

3. Substitution and the existence of the exponent

Let S° and T° be rooted binary phylogenetic trees on a label set Z. For each $z \in Z$ let A_z and B_z be rooted binary phylogenetic trees on a label set Λ_z, where the sets Λ_z are pairwise disjoint. Write $S = S^\circ[A]$ for the tree obtained from S° by identifying each leaf z with the root of A_z, and define $T = T^\circ[B]$ in the same way (Figure 1(b)). Each Λ_z is then a cluster of both S and T.

Lemma 3.1 (Substitution)**.** *With this notation,*

$$\mathrm{mast}(S,T) \le \mathrm{mast}(S^\circ,T^\circ) \cdot \max_{z \in Z} \mathrm{mast}(A_z,B_z).$$

Proof. Let Y be an agreement set for S and T, and let $Z_Y = \{z : Y \cap \Lambda_z \neq \emptyset\}$. Choose one label $y_z \in Y \cap \Lambda_z$ for each $z \in Z_Y$ and put $Y' = \{y_z : z \in Z_Y\}$. Since each Λ_z is a cluster of S met by Y' in exactly one leaf, $S|Y'$ is $S^\circ|Z_Y$ with z renamed y_z, and likewise $T|Y'$ is $T^\circ|Z_Y$ renamed. Now $S|Y' = (S|Y)|Y' \cong (T|Y)|Y' = T|Y'$, so Z_Y is an agreement set of S° and T° and $|Z_Y| \leq \mathrm{mast}(S^\circ, T^\circ)$. For each z, the restrictions of S and T to $Y \cap \Lambda_z$ are $A_z|(Y \cap \Lambda_z)$ and $B_z|(Y \cap \Lambda_z)$, and they agree, so $|Y \cap \Lambda_z| \leq \mathrm{mast}(A_z, B_z)$. Summing over $z \in Z_Y$ gives the bound. □

When all the pairs (A_z, B_z) are copies of one pair, S and T are the tree products of [5, Definition 36], and the proof above is that of [5, Lemma 39].

Corollary 3.2. *For all $h, k \geq 0$, $M(2^{h+k}) \leq M(2^h)\, M(2^k)$.*

Proof. Take S°, T° balanced of height h with $\mathrm{mast}(S^\circ, T^\circ) = M(2^h)$, and for every z take (A_z, B_z) to be a copy, relabelled onto Λ_z, of a balanced pair of height k attaining $M(2^k)$. Then S and T are balanced of height $h + k$ on the same 2^{h+k} labels, and Lemma 3.1 applies. □

For example, substituting the pair of Figure 1(a) into itself gives balanced trees on 16 leaves whose largest agreement subtree has 4 leaves, so $M(16) \leq 4$.

Proof of Theorem 1.1(1). The sequence $a_m = \log_2 M(2^m)$ is nonnegative and, by Corollary 3.2, subadditive. Fekete's lemma [4] gives $\lim a_m/m = \inf a_m/m$. □

4. The upper bound

Proof of Theorem 1.1(2). Taking $k = 3$ in [2, Theorem 4.6] gives balanced trees on $2^{11} = 2048$ leaves with maximum agreement subtrees of size $2^5 = 32$, so $M(2^{11}) \leq 2^5$. (The trees are listed in Newick format in [2, Appendix A]; we recomputed the value 32 for exactly these trees, see Section 6.) Write $m = 11j + r$ with $0 \leq r \leq 10$. Corollary 3.2 and the trivial bound $M(2^r) \leq 2^r$ give

$$M(2^m) \leq M(2^{11})^j\, M(2^r) \leq 2^{5j}\, 2^r \leq 2^{5m/11}\, 2^{10}.$$ □

Iterating the substitution gives explicit examples. For instance, substituting the 2048-leaf pair into itself produces balanced trees on 2^{22} leaves with no agreement subtree larger than 2^{10}, whereas $n^{1/2} = 2^{11}$.

Remark 4.1. The construction of [2] arranges $p = 2^{h_1}$ pendant subtrees of height h_2 in each tree as a grid, and fills each of the p^2 cells with a caterpillar that is reversed in the other tree; this uses caterpillars of length at most $h_2 + 1$, so $p(h_2 + 1) \geq 2^{h_2}$, and gives agreement $2p$ on $p\, 2^{h_2}$ leaves. Combined with substitution, one such pair yields the exponent $(1 + h_1)/(h_1 + h_2)$. Over all (h_1, h_2) allowed by the length constraint this is minimised at $(h_1, h_2) = (4, 7)$, the case $k = 3$, with value $5/11$. Going below $5/11$ therefore needs a different gadget, not a different choice of parameters.

5. The lower bound

For balanced trees S and T of heights h_1 and h_2 whose label sets share exactly $t \geq 1$ labels, let $g(h_1, h_2, t)$ be the minimum possible value of $\mathrm{mast}(S, T)$. Following [2, Theorem 1.3], an estimate of the form

$$g(h_1, h_2, t) \;\geq\; t^a\, 2^{-c(h_1+h_2)} \tag{2}$$

gives $M(n) \geq n^{a-2c}$ on putting $h_1 = h_2 = \log_2 n$ and $t = n$. The paper [2] takes $(a, c) = (0.22, 0.025)$, giving the exponent 0.17. The induction closes for any (a, c) satisfying the inequality below. For $x = (x_1, x_2, x_3, x_4)$ with nonnegative entries set

$$\Phi_{a,c}(x) = \max\Big\{4^c(x_1^a + x_4^a),\ 4^c(x_2^a + x_3^a),\ 2^c(x_1 + x_2)^a,\ 2^c(x_3 + x_4)^a,\ 2^c(x_1 + x_3)^a,\ 2^c(x_2 + x_4)^a\Big\},$$

with the convention $0^a = 0$, and let $\Delta = \{x \geq 0 : x_1 + x_2 + x_3 + x_4 = 1\}$.

In the induction, $x_1, \ldots, x_4$ are the fractions of the shared labels lying in the four overlap cells $S_p \cap T_q$, and the six terms of $\Phi_{a,c}$ are the six cases of (1): the two diagonals, where both

trees lose a level, and the two rows and two columns, where one tree does (Figure 2). The inequality says that whichever way the shared labels are spread over the cells, at least one case keeps enough of them.

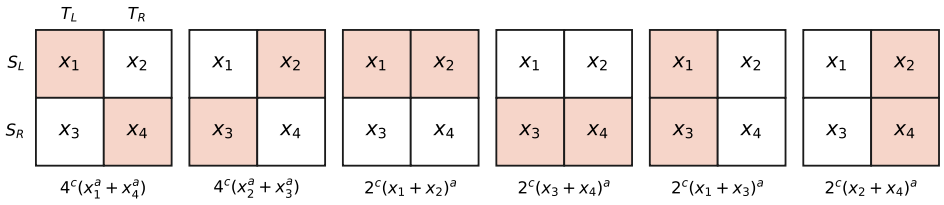


FIGURE 2. The six terms of $\Phi_{a,c}$ as the six cases of recurrence (1), read on the table of overlaps $x_1, \dots, x_4$ between the halves of S (rows) and of T (columns).

Lemma 5.1 (computer-assisted)**.** *For $a = 2/5$ and $c = 33/400$, $\Phi_{a,c}(x) \geq 1$ for every $x \in \Delta$.*

Proof. Each of the six expressions in $\Phi_{a,c}$ is nondecreasing in each coordinate. Parametrise Δ by (x_1, x_2, x_3) with $x_4 = 1 - x_1 - x_2 - x_3$. On a box $\prod_{i \leq 3}[l_i, u_i]$, every point of Δ satisfies $x_i \geq l_i$ for $i \leq 3$ and $x_4 \geq \max(0, 1 - u_1 - u_2 - u_3)$, so $\Phi_{a,c}$ is bounded below on the box by its value at $(l_1, l_2, l_3, \max(0, 1 - u_1 - u_2 - u_3))$. Starting from $[0,1]^3$, a box is discarded if $l_1 + l_2 + l_3 > 1$, accepted if this lower bound is at least 1, and otherwise bisected along its longest side. Box corners are exact dyadic rationals, a and c are the exact rationals above, and each expression is evaluated in interval arithmetic at 40 significant digits; a box is accepted only when the lower endpoint of some expression's interval is at least 1. The procedure terminates after examining 4373 boxes, the smallest of side 2^{-12}, with every box accepted or discarded. □

Proof that $M(n) \geq n^{0.235}$. We prove (2) with $(a, c) = (2/5, 33/400)$ by induction on $h_1 + h_2$. If $h_1 = 0$ or $h_2 = 0$, one tree is a single leaf, so $t = 1$ and $g = 1 \geq 2^{-c(h_1+h_2)}$. Otherwise let S_L, S_R, T_L, T_R be the maximal pendant subtrees, and let $t\,x_1$, $t\,x_2$, $t\,x_3$, $t\,x_4$ be the numbers of shared labels in $S_L \cap T_L$, $S_L \cap T_R$, $S_R \cap T_L$ and $S_R \cap T_R$, so that $x \in \Delta$. Apply the induction hypothesis to each term of (1); a term whose two trees share no labels is 0, which matches $0^a = 0$. For example

$$\operatorname{mast}(S_L, T_L) + \operatorname{mast}(S_R, T_R) \;\geq\; \big((tx_1)^a + (tx_4)^a\big) 2^{-c(h_1+h_2-2)} = t^a 2^{-c(h_1+h_2)} \cdot 4^c(x_1^a + x_4^a),$$

and $\operatorname{mast}(S_L, T) \geq (t(x_1 + x_2))^a\, 2^{-c(h_1+h_2-1)} = t^a 2^{-c(h_1+h_2)} \cdot 2^c(x_1 + x_2)^a$. The six terms give $\operatorname{mast}(S, T) \geq t^a 2^{-c(h_1+h_2)}\, \Phi_{a,c}(x)$, and Lemma 5.1 completes the induction, and gives $M(n) \geq n^{a-2c}$ with $a - 2c = 2/5 - 33/200 = 0.235$. □

Remark 5.2. Numerical optimisation over (a, c) indicates that this induction cannot give an exponent larger than about 0.2354, attained near $a = 0.40$. The binding configurations lie near $x \approx (0.73, 0.13, 0.13, 0)$: one of the four overlap cells holds most of the shared labels and the rest is split evenly across the two off-diagonal cells. Further progress on the lower bound needs more than a one-level induction.

A two-level induction. Keep the ansatz (2), but when $h_1 \geq 2$ and $h_2 \geq 1$ expand S two levels and T one level. Index the four subtrees of S at depth two by $p \in \{00, 01, 10, 11\}$ and the two subtrees of T at depth one by $q \in \{0, 1\}$, and let $t\,y_{pq}$ be the number of shared labels in $S_p \cap T_q$, so that y lies in the simplex Δ_8 of dimension seven. For a vertex u of S at depth at most two and a vertex v of T at depth at most one, write $L_{uv}(y)$ for the sum of y_{pq} over the leaves p below u and q below v, so that $t\,L_{uv}(y)$ labels are shared by S_u and T_v.

Applying (1) to the pair (S, T) and then again to pairs of subtrees, as long as both stay within the expanded levels, and stopping at pairs (S_u, T_v) other than (S, T) itself, bounds $\operatorname{mast}(S, T)$ below by a sum of terms $\operatorname{mast}(S_u, T_v)$ over disjoint pairs. Call each such choice a *strategy*;

there are 42. Every stopped pair has smaller total height, so the induction hypothesis gives $\mathrm{mast}(S_u, T_v) \geq (t\, L_{uv}(y))^a\, 2^{-c(h_1+h_2-|u|-|v|)}$, where $|u|, |v|$ are the depths. Hence

$$\mathrm{mast}(S,T) \;\geq\; t^a 2^{-c(h_1+h_2)} \max_j \varphi_j(y), \qquad \varphi_j(y) = \sum_{(u,v)\in j} 2^{c(|u|+|v|)}\, L_{uv}(y)^a.$$

Each φ_j is a nonnegative combination of concave functions of y, so it is concave on Δ_8. For example, pairing $S_0 = S_L$ with $T_0 = T_L$ and S_1 with T_1 by (1), and then using the alternative $\mathrm{mast}(S_0, T_0) \geq \mathrm{mast}(S_{00}, T_0)$ gives the strategy $\{(S_{00}, T_0), (S_1, T_1)\}$, with $\varphi_j(y) = 2^{3c} L_{00,0}(y)^a + 2^{2c} L_{1,1}(y)^a$.

Lemma 5.3 (computer-assisted)**.** *For $a = 2/5$ and $c = 157/2000$, $\max_j \varphi_j(y) \geq 1$ for every $y \in \Delta_8$.*

Proof. Subdivide Δ_8 by repeatedly bisecting the longest edge of a simplex. A simplex is certified if some φ_j, or some combination $\sum_j \lambda_j \varphi_j$ with $\lambda_j \geq 0$ and $\sum_j \lambda_j \leq 1$, is at least 1 at all of its vertices. Such a combination is concave and at most $\max_j \varphi_j$ (as each $\varphi_j \geq 0$), and a concave function attains its minimum over a simplex at a vertex, so the lemma holds on a certified simplex. The subdivision of Δ_8 into 1,418,968 certified simplices, of maximum depth 47, is found by a floating-point search and each certificate is then checked rigorously. The vertices are dyadic, and are handled as exact integers over 2^{52}, so every value L_{uv} is exact. For each power $x^{2/5}$ and each weight $2^{c(|u|+|v|)}$, a floating-point lower bound r is used only after checking $r^5 \leq x^2$, respectively $r^{2000} \leq 2^{157(|u|+|v|)}$, in exact integer arithmetic. The final sums, of at most a few dozen terms, are required to exceed $1 + 2^{-40}$, which is more than their worst-case rounding error, and the weights λ_j are checked to sum to at most 1 in exact rational arithmetic. Of the simplices, 1,042,855 are certified by a combination and the rest by a single strategy. □

Proof of Theorem 1.1(3). We prove (2) with $(a, c) = (2/5, 157/2000)$ by induction on $h_1 + h_2$. By symmetry of mast we may assume $h_1 \geq h_2$. If $h_1 \leq 1$ or $h_2 = 0$, one tree has at most two leaves, so $t \leq 2$; any set of at most two shared labels is an agreement set, so $\mathrm{mast}(S,T) = t \geq t^a \geq t^a 2^{-c(h_1+h_2)}$. Otherwise $h_1 \geq 2$ and $h_2 \geq 1$, and the display above together with Lemma 5.3 completes the induction. Putting $h_1 = h_2 = m$ and $t = n = 2^m$ gives $M(n) \geq n^{a-2c}$ with $a - 2c = 2/5 - 157/1000 = 243/1000$. □

Remark 5.4. The same certifier finds a point where the inequality fails for $c = 19/250$, so with $a = 2/5$ this induction gives no exponent above about 0.248. At $a = 2/5$, local search from that point places the threshold near $c \approx 0.0776$, an exponent of about 0.245.

6. Verification

All claims involving computation were checked with short scripts, available with the Lean files at `https://github.com/jvvk/mathematics`.

- An exact dynamic program for mast of two balanced trees, based on (1) and restricted to pairs of subtrees with overlapping label sets, agrees with brute-force enumeration on 300 random pairs with at most 16 leaves.
- Parsing the two trees printed in [2, Appendix A] confirms that both are balanced of height 11 on the labels $1, \ldots, 2048$, and the program gives $\mathrm{mast}(S,T) = 32$ for them.
- Rebuilding the construction of [2, Theorem 4.6] with $k = 3$ gives agreement exactly 32 on 2048 leaves. Substituted pairs give exactly the product bound, for example agreement 128 on 2^{15} leaves from the $k = 2$ and $k = 3$ pairs. When the caterpillars are not reversed the same program reports 128 on 2048 leaves, so it does detect a broken construction.
- The certificate of Lemma 5.1 succeeds for $(a, c) = (2/5, 33/400)$ and fails, as it should, for $(2/5, 2/25)$, which lies below the numerical threshold.
- The certificate of Lemma 5.3 succeeds for $(a, c) = (2/5, 157/2000)$ in 582 seconds and finds a violating vertex, as it should, for $(2/5, 19/250)$. Every simplex accepted by the floating-point search also passed the rigorous check. The exact lower bounds for $x^{2/5}$ were compared with 60-digit values at 20,000 random arguments and never exceeded them, and a deliberately inflated bound fails the exact test.

- The 42 strategies are generated mechanically from (1). With one level on each side the same generator and the simplicial certifier reproduce Lemma 5.1, in 331 simplices.

7. Formal verification

Everything above except Lemma 5.3 is checked in Lean 4 with Mathlib [9], using only the standard axioms. The formal development follows the text.

- Agreement is defined through displayed triples, and a separate theorem shows that this is the usual definition: Y is an agreement set exactly when $S|Y \cong T|Y$, because a rooted binary phylogenetic tree is determined by its triples.
- Lemma 3.1, Corollary 3.2 and Theorem 1.1(1) (Fekete's lemma is in Mathlib).
- Theorem 1.1(2), with $M(2^{11}) \le 2^5$ proved, not quoted: Lemma 4.5 of [2] is formalised with their proof, and applied to an explicit pair of balanced trees on 2048 leaves built as in their Theorem 4.6 from sixteen 8-caterpillars packing a tree of height 7 (found by the recursion of their Lemma 4.3).
- The one-level induction and Lemma 5.1, hence $M(n) \ge n^{0.235}$. Lean checks a second, integer certificate for Lemma 5.1: 1638 boxes on the grid of step $1/N$, $N = 2^{16}$. A lower bound $r/2^{24}$ for $x^{2/5}$ at a grid value $x = S/N$ is certified by the integer inequality $r^5N^2 \le S^2 2^{120}$, and lower bounds for 2^c and 4^c by their 400th and 200th powers. The boxes are found by a short script; Lean only checks them.
- The two-level induction with the 42 strategies, which are generated mechanically and compared with those of the certifier. Lean proves $M(n) \ge n^{0.243}$ from the inequality of Lemma 5.3, which enters as a hypothesis.

The certificate of Lemma 5.3, with over a million simplices, is checked in exact arithmetic by the script of Section 6 and not in Lean. Fourteen mutations of the formal development (a wrong constant, a caterpillar left unreversed, an inflated bound) each make it fail to compile. The Lean files and scripts are at `https://github.com/jvvk/mathematics`.

8. Open questions

The exponent β now lies in $[0.243, 5/11]$. By Theorem 1.1(1), any pair of balanced trees on 2^m leaves with agreement less than $2^{5m/11}$ improves the upper bound. Local search over labellings found no such pair with at most 64 leaves. A natural way to shorten the caterpillars in Remark 4.1 would be to use pairs of lower height in which no triple is displayed by both trees. Exhaustive enumeration shows that for $5 \le r \le 7$ every such pair on r leaves contains a tree of height $r-1$, so for these sizes the caterpillar is already optimal. Whether β is close to $5/11$ or substantially smaller is open.

Changes in version 3. The substitution argument is credited to Guillemot, Nicolas, Berry and Paul, and the version of [2] whose numbering is cited is stated. The verification programs now include the unreversed-caterpillar control and the log of the full run for Lemma 5.3, and the mutation tests count a timeout as a failure of the test. The theorems, their proofs and the Lean statements are unchanged.

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